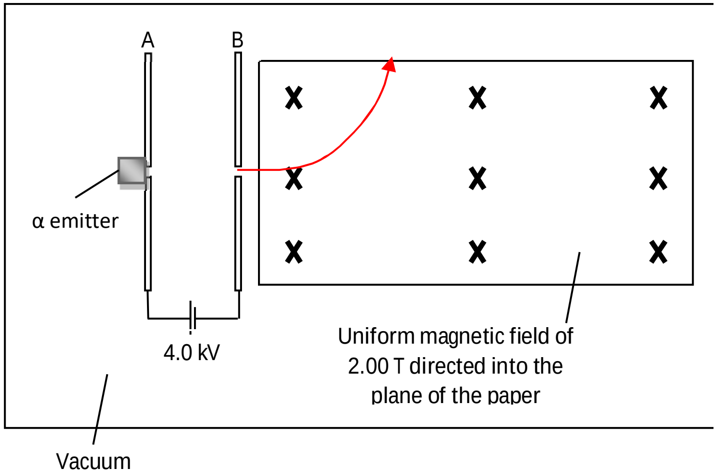
affected by external factor

8 (a)

One tesla is defined as the amount of magnetic flux density of a uniform magnetic field when a magnetic force per unit current per unit length of 1 newton per ampere per metre acts on a straight wire placed perpendicular to the magnetic field.

8 (b)(i)

Award [B1] if student's answer shows evidence of a circular path and particle is deflected upwards.



8 (b)(ii)

The magnetic force acting on the α-particle is always directed at right angles to the velocity of the particle. [B1]

The resulting acceleration is directed at right angles to the velocity of the particle and thus will not alter the speed but merely changes its direction. [B1] **OR** The magnetic force hence does not do work on the particle. [B1]

The kinetic energy of the particle thus does not change. [B1]

8 (b)(iii)

$$\frac{1}{2}mv^2 = qV$$

$$v = \sqrt{\frac{2\left(2\times1.6\times10^{-19}\right)(4000)}{6.644\times10^{-27}}} \quad \text{[M1]}$$

$$= 6.207\times10^{5}\ \text{m s}^{-1} \quad \text{[A0]}$$

8 (b)(iv)

$$F_\text{B} = F_\text{E}$$

$$qvB = qE$$

$$E = vB$$

$$= (6.207\times10^{5})(2.00) \quad \text{[M1]}$$

$$= 1.24\times10^{6}\ \text{V m}^{-1} \quad \text{[A1]}$$

E Field is directed downwards [B1]

8 (c)(i)

$$F_\text{B} = F_\text{c}$$

$$qvB = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB}$$

$$= \frac{6.644\times10^{-27}\left(6.207\times10^{5}\right)\sin30^\circ}{\left(2\times1.6\times10^{-19}\right)(2.00)} \quad \text{[M1]}$$

$$= 3.22\times10^{-3}\ \text{m} \quad \text{[A1]}$$

8 (c)(ii)

Magnetic force on charged particle provides the centripetal force.

$$q_\alpha v_y B = m_\alpha v_y \omega$$

$$q_\alpha B = m_\alpha\left(\frac{2\pi}{T}\right) \quad \text{[M1]}$$

$$T = \frac{2\pi m_\alpha}{q_\alpha B} \quad \text{(shown)}$$

8 (c)(iii)

$$T = \frac{2\pi m_\alpha}{q_\alpha B}$$
$$= \frac{2\pi\left(6.644\times10^{-27}\right)}{\left(2\times1.6\times10^{-19}\right)(2.00)}$$
$$= 6.522\times10^{-8}\text{ s} \qquad [M1]$$

Pitch, $p = v_x T$

$= \left(6.207\times10^{5}\right)\cos 30^\circ \times 6.522\times10^{-8}$ [M1]

$= 0.0351\text{ m}$ [A1]

8 (c)(iv)

Since $q_{\text{positron}} = \frac{1}{2}q_\alpha$, $m_{\text{positron}} << m_\alpha$

$$\frac{m_{\text{positron}}}{q_{\text{positron}}} = \frac{9.11\times10^{-31}}{1.6\times10^{-19}} = 5.69\times10^{-12}$$
$$\frac{m_{\text{alpha}}}{q_{\text{alpha}}} = \frac{6.644\times10^{-27}}{2\times1.6\times10^{-19}} = 2.07\times10^{-8}$$

Thus, $(m/q)_{\text{positron}} < (m/q)_{\text{alpha}}$ [M1]

For a positron, the period, $T = \frac{2\pi m}{qB}$ will decrease for the same magnetic field [A1]

Since radius is also proportional to the ratio (m/q) for the same B and speed, the radius $r = \frac{mv}{qB}$ for positron will also decrease. [A1]

Pitch is proportional to the period for the same horizontal speed $p = v_x T$, thus the pitch will decrease. [A1]

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9 (a)(i)